

Chapter 24: Fishes

1. What Is a Fish?

- Definition: An aquatic vertebrate with gills, appendages (if present) in the form of fins, and usually a skin with scales of dermal origin.
- Taxonomy: “Fish” is not a monophyletic group (paraphyletic) because the ancestor of land vertebrates (tetrapods) is found within one group of fishes (sarcopterygians).
- Diversity: Estimated 28,000 living species, more than all other vertebrates combined.
- Adaptations:
 - Neutral buoyancy (swim bladder).
 - Gills for extracting oxygen.
 - Excellent olfactory and visual senses.
 - Lateral-line system for detecting water currents and vibrations.

2. Ancestry and Relationships

- Origin: Descended from an unknown free-swimming protochordate ancestor about 550 million years ago.
- Earliest Vertebrates: Jawless agnathan fishes (ostracoderms).
- Gnathostomes: Jawed fishes.
 - Placoderms: Extinct, heavy dermal armor.
 - Chondrichthyes: Cartilaginous fishes (sharks, rays, chimaeras). Lost heavy armor, adopted cartilage.
 - Acanthodians: Extinct, heavy spines on fins. Sister group of bony fishes.
 - Osteichthyes (Bony Fishes):
 - Actinopterygii: Ray-finned fishes (most modern fishes).
 - Sarcopterygii: Lobe-finned fishes (lungfishes, coelacanths, ancestors of tetrapods).

3. Living Jawless Fishes (Agnatha)

- Class Myxini: Hagfishes (盲鳗纲)
 - Marine scavengers and predators.
 - Almost blind, keen sense of smell and touch.
 - Rasping tongue with keratinized teeth.
 - Produce enormous quantities of slime.
 - Body fluids isosmotic with seawater.
 - No vertebrae, but have a cranium.
- Class Petromyzontida: Lampreys (七鳃鳗纲)
 - Marine and freshwater.
 - Parasitic (feed on body fluids of fish) or nonparasitic (do not feed as adults).
 - Anadromous (溯河洄游): Ascend freshwater streams to breed.
 - Ammocoetes (沙隐虫): Larval stage, resembles amphioxus, suspension feeder.
 - Invasion of Great Lakes by sea lamprey caused collapse of lake trout fishery.

4. Class Chondrichthyes: Cartilaginous Fishes (软骨鱼纲)

- Characteristics:
 - Cartilaginous skeleton (bone absent, derived feature).
 - Placoid scales (盾鳞).
 - No swim bladder (large liver with squalene for buoyancy).
 - Intestine with spiral valve (螺旋瓣).
 - Internal fertilization (claspers in males).

- Heterocercal tail (歪尾).
- Subclass Elasmobranchii (板鳃亚纲): Sharks, Skates, Rays.
 - Sharks: Fusiform body, predatory. Sensory systems: olfactory, lateral-line (vibration), vision, ampullae of Lorenzini (electroreception).
 - Skates and Rays: Dorsoventrally flattened, enlarged pectoral fins, bottom dwellers. Electric rays (electric organs), Stingrays (venomous spines).
- Subclass Holocephali (全头亚纲): Chimaeras (ratfish).
 - Jaws with large flat plates (upper jaw fused to cranium).
 - Operculum covers gill slits.

5. Osteichthyes: Bony Fishes (硬骨鱼)

- Characteristics: Endochondral bone (软骨内成骨), lung or swim bladder, bony operculum.
- Class Actinopterygii: Ray-Finned Fishes (辐鳍鱼纲)
 - Palaeoniscids: Earliest forms, ganoid scales (硬鳞), heterocercal tail.
 - Cladistia: Bichirs (多鳍鱼).
 - Chondrosteans: Sturgeons (鲟), Paddlefishes (匙吻鲟).
 - Neopterygians:
 - Early forms: Bowfin (*Amia*), Gars (*Lepisosteus*).
 - Teleosts (真骨鱼): Modern bony fishes (96% of living fishes).
 - Cycloid (圆鳞) or ctenoid (栉鳞) scales (light, flexible).
 - Homocercal tail (正尾).
 - Suction feeding (jaw protrusion).
- Class Sarcopterygii: Lobe-Finned Fishes (肉鳍鱼纲)
 - Ancestor of tetrapods (Rhipidistians).
 - Diphyccercal tail (原尾/矛尾).
 - Fleshy, paired lobed fins.
 - Lungfishes (肺鱼): *Neoceratodus* (Australia), *Lepidosiren* (S. America), *Protopterus* (Africa).
 - Coelacanth (腔棘鱼): *Latimeria*, thought extinct, rediscovered in 1938.

6. Structural and Functional Adaptations

- Locomotion:
 - Myomeres (W-shaped muscle bands).
 - Undulatory movement (reactive force = thrust + lateral force).
 - Fast swimmers (tuna) use stiff caudal fin, regional endothermy (warm muscles).
- Neutral Buoyancy:
 - Swim Bladder (鳔): Gas-filled space to adjust density.
 - Physostomous (鳔管鱼类): Pneumatic duct connects to esophagus.
 - Physoclistous (闭鳔鱼类): No duct; gas secreted by gas gland (rete mirabile, countercurrent multiplier) and resorbed by ovale.
- Hearing:
 - Weberian Ossicles (韦伯氏器): Bones transmitting vibrations from swim bladder to inner ear (Ostariophysans: minnows, catfish, etc.).
- Respiration:
 - Gills: Filaments with lamellae, countercurrent flow (blood vs. water) maximizes oxygen extraction.
 - Ram ventilation (tuna, mackerel).
 - Air breathing (lungfish, electric eel, climbing perch).
- Osmotic Regulation:

- ▶ Freshwater Fishes (Hyperosmotic regulators): Water enters, salt lost.
 - Kidneys pump excess water (dilute urine).
 - Salt-absorbing cells in gills.
- ▶ Marine Fishes (Hypoosmotic regulators): Water lost, salt enters.
 - Drink seawater.
 - Salt-secretory cells in gills secrete NaCl.
 - Kidneys excrete divalent ions (Mg, SO₄).
- Feeding: Carnivores, herbivores, suspension feeders (gill rakers), scavengers, parasites.
- Migration:
 - ▶ Catadromous (降河洄游): Freshwater to sea to spawn (e.g., Freshwater Eels *Anguilla* - spawn in Sargasso Sea).
 - ▶ Anadromous (溯河洄游): Sea to freshwater to spawn (e.g., Salmon - homing instinct via odor imprinting).
- Reproduction:
 - ▶ Mostly dioecious, external fertilization (oviparous).
 - ▶ Some ovoviviparous (guppies) or viviparous.
 - ▶ Parental care: Variable (pelagic spawners = no care; benthic spawners = often guard eggs).
 - ▶ Growth: Temperature dependent (annual rings on scales/otoliths).